CS & IT ENGINEERING

2024

Discrete Mathematics

Set Theory & Algebra

DPP – 06 Discussion Notes



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[MSQ]

#Q. The set of all positive rational numbers forms an abelian group under the composition * defined by a * b = (ab)/2.

Which of the following is/are TRUE ?

A. The identity element is 2

B. The inverse of a is 4/a.

C. The inverse of 4 is 1

D. The identity element is 1

$a * b = \frac{ab}{2}$

$a * 2 = \frac{a \times 2}{2}$

$a * b = 2$

$\Rightarrow \frac{ab}{2} = 2$

$\Rightarrow b = 4/a$

$a = 4$

$\frac{4}{a} = \frac{4}{4} = 1$

[MSQ]

#Q. Let r be the set of all real numbers and * is a binary operation defined by $a * b = a + b + ab$.

Which of the following is TRUE ?

A Identity element is 0.

B The inverse of –1 is 1.

C The inverse of a is $-a/(a + 1)$.

D R is not a group.

$a * 0 = 0 * a = a$

$a + 0 + a \times 0 = a$

$0 + a + 0 \times a = a$

$-1 * 1 = -1 + 1 + (-1 \times 1) = -1 \neq 0$

$a * b = 0$

$\Rightarrow a + b + ab = 0$

$\Rightarrow b(1 + a) = -a$

$\Rightarrow b = \frac{-a}{1+a}$

[MCQ]

#Q. The set G = {0, 1, 2, 3, 4, 5} is a group with respect to addition modulo 6. Which of the following is false ?

$e = 0$

$2 \oplus_6 4 = 0$

A The inverse of 2 is 4

B The inverse of 3 is 3

C The inverse of 5 is 2 $= 7 \% 6 = 1$

D The inverse of 1 is 5

[NAT]

#Q. G = {1, –1, i, – i}is a group w.r.t multiplication. The order –i is

$e = 1$

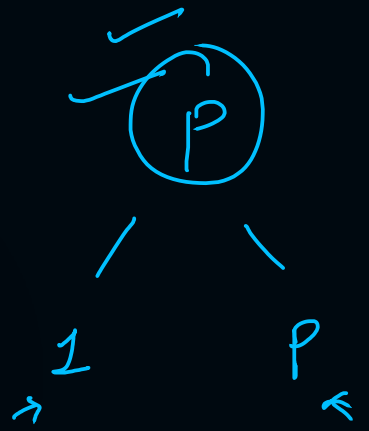
$(-i)^n = 1$

$(-i)^4 = 1$

4

[MCQ]

#Q. If G is a group of order p, where p is a prime number. Then the number of sub groups of G is___.



A 1

B 2

C p – 1

D p

[MSQ]

#Q. Which of the following is True ?

$G_4 = \{1, 3, 5, 7\}$ $\otimes_8$

$H_1 = \{1, 3\}$

$H_2 = \{1, 5\}$

$H_1 \cup H_2 = \{1, 3, 5\}$

$15 \% 8 = 7$

A. The union of any two sub groups of a group G is also a sub group of G.

B. The intersection of any two sub groups of a groups of a group G is also a sub group of G.

C. Let G = {0, ± k, ± 2k, ± 3k, ± 4k, ...∞} where k is any fixed integer is a group w.r.t. addition.

D. Every sub group of an abelian group is also an abelian group.

[MSQ]

#Q. Which of the following is True ?

A every cyclic group is an abelian group

B if ‘a’ is generator of a cyclic group then a–1 is also a generator of G.

C The order of a cyclic group is equal to the order of its generating element.

D A sub group of a cyclic group need not be cyclic.

[NAT]

#Q. How many generators are there for the cyclic group $(\{1, 2, 3, 4, 5, 6\}, \oplus 7)$

$$\phi(6) = \phi(2 \times 3) = \phi(2) \times \phi(3)$$

$$= (2-1)(3-1)$$

$$= 2$$

[MCQ]

#Q. Which of the following is false ?

A. A cyclic group with only one generator can have at most 2 elements.

B. The order of a cyclic group is equal to the order of its generator.

C. The group $(\{1, 2, 3, 4\}, \otimes 5)$ is cyclic.

$\{1, 3, 5, 7\}$ $\otimes_8$

D. A group of order 4 is cyclic.

[MCQ]

#Q. Let S = {0, 1, 2, 3, 4, 5, 6, 7} and * denotes multiplication modulo 8 that is $x * y = (x\,y) \bmod 8$.

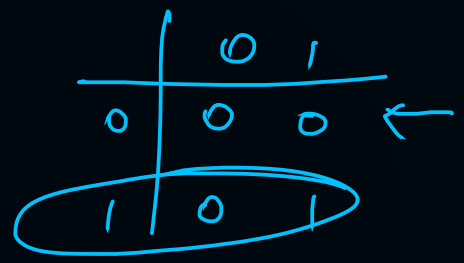
Which of the following is a group w.r.t * ?

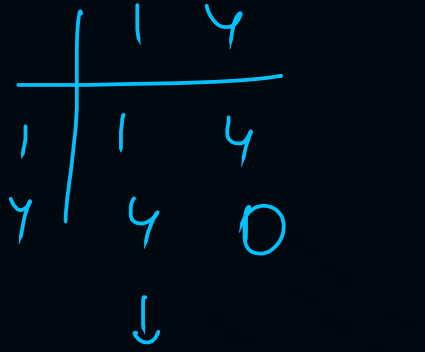
A {0, 1}

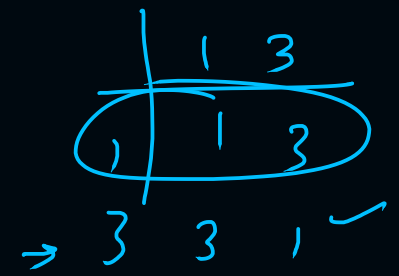
B {1, 4}

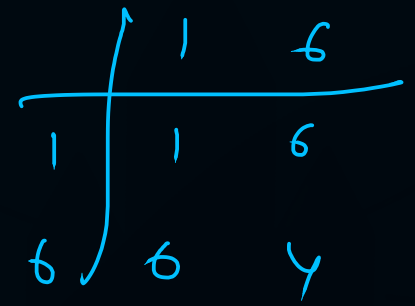
C {1, 3}

D {1, 6}









THANK - YOU